Electromagnetism Test Document for VimTeX, SyncTeX, LSP, and Snippets

Generated Test File

June 28, 2026

Abstract

This document is a deliberately equation-heavy electromagnetism sample for testing a LaTeX editing setup. It includes cross-references, citations, displayed equations, aligned environments, a bibliography, TikZ figures, tables, labels, and enough page breaks to make forward and inverse SyncTeX testing meaningful. Compile with `latexmk -pdf -synctex=1 em-test.tex`. For VimTeX, try `\ll` to compile, `\lv` to view, `\le` for errors, and PDF-to-source inverse search in your viewer.

Contents

1 Electrostatics and Coulomb's Law

Electrostatics begins with the empirical observation that stationary charges exert forces on one another. For point charges q_1 and q_2 separated by the vector $\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$, Coulomb's law is

$$\boldsymbol{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{\|\boldsymbol{r}\|^2}\hat{\boldsymbol{r}}. \tag{1}$$

This is useful for force calculations, but field language is usually cleaner. The electric field due to a point charge q at the origin is

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}. \tag{2}$$

For a continuous charge distribution $\rho(\boldsymbol{r}')$, the superposition principle gives

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\int_{\mathbb{R}^3}\rho(\boldsymbol{r}')\frac{\boldsymbol{r}-\boldsymbol{r}'}{\|\boldsymbol{r}-\boldsymbol{r}'\|^3}\,\mathrm{d}^3 r'. \tag{3}$$

A useful test for editor navigation is to jump between equation (1), equation (2), and equation (3). The conceptual move from forces to fields is emphasized in standard treatments such as [1], [2].

1.1 Potential energy and electric potential

Because the electrostatic field is conservative, it may be written as the negative gradient of a scalar potential:

$$\boldsymbol{E} = -\boldsymbol{\nabla} V. \tag{4}$$

The potential generated by a charge distribution is

$$V(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\boldsymbol{r}')}{\|\boldsymbol{r}-\boldsymbol{r}'\|}\,\mathrm{d}^3 r'. \tag{5}$$

This expression is often easier to evaluate than the vector integral in equation (3), after which the field can be recovered from equation (4).

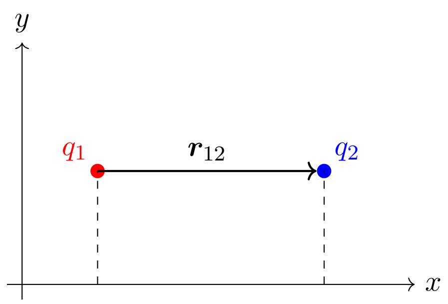

Figure 1: Two point charges separated by a displacement vector. Use this figure to test SyncTeX jumps near captions, equations, and text.

See figure 1 and test whether clicking near the caption in the PDF jumps to this source location.

2 Gauss's Law

Gauss's law is a compact expression of the relationship between electric flux and enclosed charge. In integral form,

$$\oint_{\partial\Omega} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{a} = \frac{Q_{\mathrm{enc}}}{\varepsilon_0}. \tag{6}$$

Using the divergence theorem,

$$\oint_{\partial\Omega} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{a} = \int_{\Omega} \boldsymbol{\nabla} \cdot \boldsymbol{E} \, \mathrm{d}^3 r, \tag{7}$$

and since $Q_{\mathrm{enc}} = \int_{\Omega} \rho \, \mathrm{d}^3 r$, one obtains the differential form

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0}. \tag{8}$$

2.1 Spherical symmetry

For a point charge or uniformly charged sphere, symmetry forces the field to be radial:

$$\boldsymbol{E}(\boldsymbol{r}) = E(r)\hat{\boldsymbol{r}}. \tag{9}$$

A spherical Gaussian surface gives

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{a} = E(r) 4\pi r^2, \tag{10}$$

$$E(r) 4\pi r^2 = \frac{q}{\varepsilon_0}, \tag{11}$$

$$E(r) = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2}. \tag{12}$$

This recovers equation (2), but with far less calculation.

2.2 Planar and cylindrical symmetry

For an infinite charged sheet with surface charge density σ, a pillbox Gaussian surface gives

$$E = \frac{\sigma}{2\varepsilon_0}. \tag{13}$$

For an infinite line charge with linear density λ, a cylindrical Gaussian surface gives

$$E(s) = \frac{\lambda}{2\pi\varepsilon_0 s}. \tag{14}$$

These examples are good places to test text-object motions, surrounding delimiters, and snippets for equations.

Table 1: Common Gaussian surfaces and resulting electric fields.

Source	Gaussian surface	Field magnitude
Point charge q	sphere	$q/(4\pi\varepsilon_0 r^2)$
Infinite line λ	cylinder	$\lambda/(2\pi\varepsilon_0 s)$
Infinite sheet σ	pillbox	$\sigma/(2\varepsilon_0)$

Table 1 is included to test table navigation, captions, and references.

3 Boundary Conditions and Conductors

In electrostatic equilibrium, the electric field inside a conductor is zero:

$$\boldsymbol{E}_{\text{inside}} = \boldsymbol{0}. \tag{15}$$

If this were not true, free charges would accelerate until they rearranged themselves to cancel the internal field.

The tangential component of the electrostatic field is continuous across a boundary:

$$\hat{\boldsymbol{n}} \times (\boldsymbol{E}_2 - \boldsymbol{E}_1) = \boldsymbol{0}. \tag{16}$$

The normal component jumps by an amount determined by the surface charge density:

$$\hat{\boldsymbol{n}} \cdot (\boldsymbol{E}_2 - \boldsymbol{E}_1) = \frac{\sigma}{\varepsilon_0}. \tag{17}$$

Together, equations (16) and (17) are among the most useful local constraints in electrostatics.

3.1 Capacitance

For two conductors carrying charges $+Q$ and $-Q$, the capacitance is

$$C = \frac{Q}{\Delta V}. \tag{18}$$

For a parallel-plate capacitor with plate area A and separation d, neglecting fringing fields,

$$C = \frac{\varepsilon_0 A}{d}. \tag{19}$$

The stored electrostatic energy can be written as

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV. \tag{20}$$

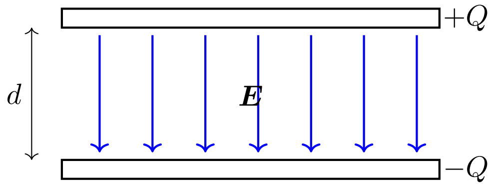

Figure 2: Idealized parallel-plate capacitor. The field is approximately uniform between the plates.

Try jumping from equation (19) to figure 2 using your LSP or Telescope symbol tools.

4 Poisson and Laplace Equations

Combining $\boldsymbol{E} = -\boldsymbol{\nabla} V$ with Gauss's law yields Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}. \tag{21}$$

In regions with no charge density, this reduces to Laplace's equation:

$$\nabla^2 V = 0. \tag{22}$$

4.1 Uniqueness

A central result is that solutions to Laplace's equation are unique once the boundary values are specified. Suppose V_1 and V_2 both solve Laplace's equation in a region Ω and agree on $\partial\Omega$. Define $U = V_1 - V_2$. Then

$$\nabla^2 U = 0, \qquad U|_{\partial\Omega} = 0. \tag{23}$$

Using Green's first identity,

$$\int_\Omega \|\boldsymbol{\nabla} U\|^2 \, \mathrm{d}^3 r = \oint_{\partial\Omega} U \boldsymbol{\nabla} U \cdot \mathrm{d}\boldsymbol{a} - \int_\Omega U \nabla^2 U \, \mathrm{d}^3 r = 0. \tag{24}$$

Therefore $\boldsymbol{\nabla} U = \mathbf{0}$, and the boundary condition forces $U = 0$, so $V_1 = V_2$.

4.2 Separation of variables

In Cartesian coordinates,

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{25}$$

For a rectangular region, one may try

$$V(x, y, z) = X(x)Y(y)Z(z), \tag{26}$$

which turns the partial differential equation into ordinary differential equations. This section is useful for testing snippets involving partial derivatives and aligned equations.

$$\frac{1}{X}\frac{\mathrm{d}^2 X}{\mathrm{d}x^2} + \frac{1}{Y}\frac{\mathrm{d}^2 Y}{\mathrm{d}y^2} + \frac{1}{Z}\frac{\mathrm{d}^2 Z}{\mathrm{d}z^2} = 0. \tag{27}$$

5 Magnetostatics

Magnetostatics describes steady currents and time-independent magnetic fields. The Lorentz force on a charge q moving with velocity $\boldsymbol{v}$ is

$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}). \tag{28}$$

For a current distribution $\boldsymbol{J}(\boldsymbol{r}')$, the Biot–Savart law gives

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \int \boldsymbol{J}(\boldsymbol{r}') \times \frac{\boldsymbol{r} - \boldsymbol{r}'}{\|\boldsymbol{r} - \boldsymbol{r}'\|^3} \, \mathrm{d}^3 r'. \tag{29}$$

5.1 Ampere's law

For steady currents, Ampere's law is

$$\oint_{\partial S} \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{\ell} = \mu_0 I_{\text{enc}}. \tag{30}$$

Using Stokes' theorem gives

$$\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}. \tag{31}$$

The magnetic field of a long straight wire follows immediately:

$$B(s) = \frac{\mu_0 I}{2\pi s}. \tag{32}$$

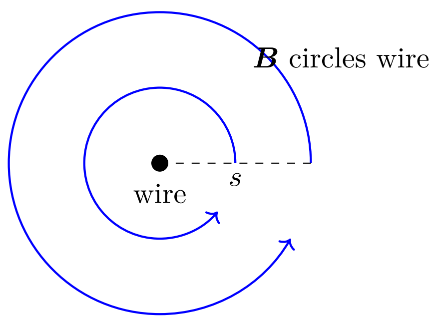

Figure 3: Magnetic field lines around a long straight current-carrying wire.

Magnetostatic calculations often mirror electrostatic ones, but with curls replacing divergences. Compare equation (8) with equation (31).

6 Vector Potential and Gauge Freedom

Because magnetic fields are solenoidal,

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0, \tag{33}$$

one may define a vector potential $\boldsymbol{A}$ such that

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{34}$$

The vector potential is not unique. If χ is a scalar function, then

$$\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\chi \tag{35}$$

produces the same magnetic field because

$$\boldsymbol{\nabla} \times \boldsymbol{A}' = \boldsymbol{\nabla} \times \boldsymbol{A} + \boldsymbol{\nabla} \times \boldsymbol{\nabla}\chi = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{36}$$

6.1 Coulomb gauge

One common choice is the Coulomb gauge,

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0. \tag{37}$$

In magnetostatics, combining equations (31), (34) and (37) yields

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}, \tag{38}$$

$$-\nabla^2 \boldsymbol{A} = \mu_0 \boldsymbol{J}, \tag{39}$$

$$\nabla^2 \boldsymbol{A} = -\mu_0 \boldsymbol{J}. \tag{40}$$

This resembles Poisson's equation in equation (21), but applies component-wise to the vector potential.

6.2 Aharonov–Bohm intuition

Even where $\boldsymbol{B} = \mathbf{0}$, the vector potential may influence quantum phases. This is one reason the vector potential is more than a mere computational convenience in quantum mechanics. A classical electromagnetism course may introduce this only briefly, while more advanced treatments discuss it in detail [3].

7 Time-Dependent Fields and Maxwell's Correction

The magnetostatic form of Ampere's law is incomplete for time-dependent situations. Taking the divergence of $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}$ gives

$$0 = \mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{J}, \tag{41}$$

which would require charge conservation in the restricted form $\boldsymbol{\nabla} \cdot \boldsymbol{J} = 0$. But the continuity equation is

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{J} = 0. \tag{42}$$

Using Gauss's law,

$$\rho = \varepsilon_0 \boldsymbol{\nabla} \cdot \boldsymbol{E}, \tag{43}$$

the continuity equation becomes

$$\boldsymbol{\nabla} \cdot \left(\boldsymbol{J} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right) = 0. \tag{44}$$

Maxwell's corrected Ampere law is therefore

$$\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}. \tag{45}$$

7.1 Faraday's law

Faraday's law relates changing magnetic flux to induced electric fields:

$$\oint_{\partial S} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{\ell} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{a}. \tag{46}$$

In differential form,

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}. \tag{47}$$

This is a good place to test diagnostics by intentionally deleting a brace, then using `\le` to inspect errors.

8 Maxwell's Equations

In vacuum with sources, Maxwell's equations are

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0}, \tag{48}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{49}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{50}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}. \tag{51}$$

Together these equations unify electrostatics, magnetostatics, induction, and electromagnetic waves. The source-free case is obtained by setting $\rho = 0$ and $\boldsymbol{J} = \boldsymbol{0}$.

8.1 Constitutive relations

In matter, it is often more useful to introduce $\boldsymbol{D}$ and $\boldsymbol{H}$:

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P}, \tag{52}$$

$$\boldsymbol{H} = \frac{1}{\mu_0}\boldsymbol{B} - \boldsymbol{M}. \tag{53}$$

For simple linear media,

$$\boldsymbol{D} = \varepsilon \boldsymbol{E}, \tag{54}$$

$$\boldsymbol{B} = \mu \boldsymbol{H}. \tag{55}$$

Then the source equations become

$$\nabla \cdot \boldsymbol{D} = \rho_{\text{free}}, \tag{56}$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{J}_{\text{free}} + \frac{\partial \boldsymbol{D}}{\partial t}. \tag{57}$$

Table 2: Differential Maxwell equations in vacuum.

Equation	Meaning
$\nabla \cdot \boldsymbol{E} = \rho/\varepsilon_0$	electric charge sources electric field
$\nabla \cdot \boldsymbol{B} = 0$	no magnetic monopoles in classical EM
$\nabla \times \boldsymbol{E} = -\partial \boldsymbol{B}/\partial t$	changing magnetic field induces electric circulation
$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \partial \boldsymbol{E}/\partial t$	current and displacement current source magnetic circulation

Use table 2 to test citation, table, and reference completion.

9 Electromagnetic Waves

In vacuum with no charges or currents, Maxwell's equations imply wave equations for both fields. Taking the curl of Faraday's law gives

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{B}), \tag{58}$$

$$\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2}. \tag{59}$$

Since $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$, this becomes

$$\nabla^2 \boldsymbol{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} = 0. \tag{60}$$

Similarly,

$$\nabla^2 \boldsymbol{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{B}}{\partial t^2} = 0. \tag{61}$$

The wave speed is

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}. \tag{62}$$

9.1 Plane waves

A monochromatic plane wave propagating in the $+\hat{\boldsymbol{z}}$ direction may be written

$$\boldsymbol{E}(z,t) = E_0 \cos(kz - \omega t)\hat{\boldsymbol{x}}, \tag{63}$$

$$\boldsymbol{B}(z,t) = \frac{E_0}{c} \cos(kz - \omega t)\hat{\boldsymbol{y}}. \tag{64}$$

The fields are transverse, mutually perpendicular, and related by

$$\boldsymbol{B} = \frac{1}{c}\hat{\boldsymbol{k}} \times \boldsymbol{E}. \tag{65}$$

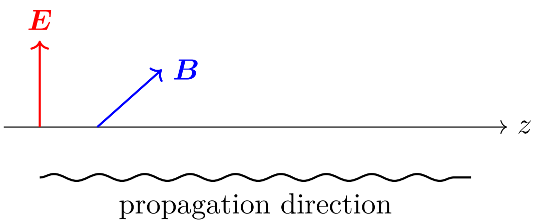

Figure 4: Schematic transverse electromagnetic plane wave.

The derivation of equations (60) and (61) is a useful stress test for equation alignment, references, and SyncTeX movement.

10 Energy, Momentum, and the Poynting Vector

The electromagnetic energy density in vacuum is

$$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2. \tag{66}$$

The Poynting vector is

$$\boldsymbol{S} = \frac{1}{\mu_0} \boldsymbol{E} \times \boldsymbol{B}. \tag{67}$$

It represents electromagnetic energy flux.

10.1 Poynting theorem

Starting from Maxwell's equations, one can derive

$$\frac{\partial u}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{S} = -\boldsymbol{J} \cdot \boldsymbol{E}. \tag{68}$$

The term $\boldsymbol{J} \cdot \boldsymbol{E}$ is the rate per unit volume at which fields do work on charges. If $\boldsymbol{J} = \boldsymbol{0}$, then electromagnetic energy is locally conserved:

$$\frac{\partial u}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{S} = 0. \tag{69}$$

10.2 Radiation pressure

Electromagnetic waves also carry momentum. For a wave normally incident on a perfectly absorbing surface, the radiation pressure is

$$p = \frac{\langle S \rangle}{c}. \tag{70}$$

For a perfectly reflecting surface, the pressure doubles:

$$p = \frac{2\langle S \rangle}{c}. \tag{71}$$

This section is good for testing references to late-document equations such as equations (67) and (68).

11 Potentials, Retardation, and a Final Checklist

For time-dependent fields, the scalar and vector potentials are often written

$$\boldsymbol{E} = -\boldsymbol{\nabla} V - \frac{\partial \boldsymbol{A}}{\partial t}, \tag{72}$$
$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{73}$$

In the Lorenz gauge,

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} + \mu_0 \varepsilon_0 \frac{\partial V}{\partial t} = 0, \tag{74}$$

the potentials satisfy wave equations:

$$\nabla^2 V - \mu_0 \varepsilon_0 \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\varepsilon_0}, \tag{75}$$
$$\nabla^2 \boldsymbol{A} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{A}}{\partial t^2} = -\mu_0 \boldsymbol{J}. \tag{76}$$

Their solutions are retarded potentials:

$$V(\boldsymbol{r}, t) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\boldsymbol{r}', t_r)}{\|\boldsymbol{r} - \boldsymbol{r}'\|} \, \mathrm{d}^3 r', \tag{77}$$
$$\boldsymbol{A}(\boldsymbol{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{r}', t_r)}{\|\boldsymbol{r} - \boldsymbol{r}'\|} \, \mathrm{d}^3 r', \tag{78}$$

where

$$t_r = t - \frac{\|\boldsymbol{r} - \boldsymbol{r}'\|}{c}. \tag{79}$$

11.1 Editing setup checklist

Use this document to test the following workflow items:

1. Compile continuously with `\ll`.
2. Open or forward-search the PDF with `\lv`.
3. Jump from PDF back to source using inverse SyncTeX.
4. Inspect LaTeX errors with `\le`.
5. Open VimTeX information with `\li` or `:VimtexInfo`.
6. Search symbols such as `Maxwell` and `Poynting` with Telescope or LSP document symbols.
7. Test snippets inside math zones, for example fractions, hats, bars, bras, kets, and display equations.
8. Test citations such as [1], [2], and [3].

11.2 Deliberate navigation targets

Here are some source locations that should be easy to find from the PDF:

- Equation (1): Coulomb's law.
- Equation (8): Gauss's law in differential form.
- Equation (29): Biot–Savart law.
- Equation (51): Maxwell–Ampere law.
- Equation (60): electromagnetic wave equation.
- Equation (67): Poynting vector.
- Equation (79): retarded time.

References

[1] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. Cambridge University Press, 2017.

[2] E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, 3rd ed. Cambridge University Press, 2013.

[3] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. Wiley, 1999.